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ARIA-NBV: Target-Conditioned, Quality-Driven Next-Best-View Planning

ARIA-NBV: Zielkonditionierte, qualitaetsgetriebene Next-Best-View-Planung

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Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the proposal text. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

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1 Problem and Research Contract

ARIA-NBV studies target-conditioned NBV for egocentric indoor reconstruction. At each decision step, an agent has a partial ASE trajectory, accumulated semi-dense or rendered geometry, frozen logged EFM/EVL context, a selected target, and a finite set of candidate camera poses. The privileged ground-truth mesh and GT target crop are oracle and evaluation assets only. The main actor-visible protocol is OBS-SEL / PRED-Q / GT-EVAL: select a target from observed or predicted evidence, condition scoring and planning on the predicted/observed target record, and evaluate with matched GT target RRI.

The thesis question is whether this setup can support target-conditioned, RRI-based multi-step view selection rather than only one-step scoring. The minimal quantitative success bar is therefore not a continuous policy. It is a discrete finite-candidate Q_H model that predicts bounded cumulative target RRI and whose selected actions are re-evaluated by the oracle. If learned Q_H does not beat one-step greedy or one-step model scoring under the same acquisition budget, the thesis must report the blocker rather than reclassifying Q_H as optional. The boundary is fixed: ASE snippets, GT meshes for oracle labels, EVL/EFM context, discrete candidates, and Rerun inspection are in scope; GT OBBs/crops are V0 sanity or GT-EVAL labels, not main actor input; Gumbel-Top-k, IQL, actor-critic, online simulators, and real-device runs are bridge work.

The literature role is correspondingly narrow. VIN-NBV provides the quality-driven RRI candidate-ranking precedent [Fra+25]. ASE and EFM3D provide the egocentric synthetic data and frozen representation substrate [Met25, Str+24]. GenNBV and Hestia show why continuous and hierarchical NBV are important, but they rely on simulator/control assumptions that ARIA-NBV should not claim before the finite-candidate result exists [Che+24, Lu+26]. Offline RL references such as Double DQN, CQL, BCQ, Decision Transformer, and IQL guide value-learning and sequence-modeling baselines, but the first thesis path is a masked finite-candidate Q_H trained from ASE oracle traces [Che+21, FMP19, HGS15, KNL21, Kum+20].

2 Objectives

The thesis objective is a reproducible target-aware finite-candidate NBV stack for egocentric reconstruction. The evidence gates are: a passable M1 offline-store/oracle contract; inspectable target RRI under V0 sanity and V1 OBS-SEL / PRED-Q / GT-EVAL; a target-conditioned one-step scorer evaluated by held-out ranking, top-k oracle hit, calibration, and failure cases; and candidate-query Q_H trained from random-valid, oracle-greedy/lookahead, and oracle-scored temperature-softmax traces. The exit condition is oracle-evaluated Q_H actions beating one-step greedy/model scoring under equal acquisition budget, or a documented blocker.

The required first target input is observed or predicted OBB geometry plus class, confidence, projected area, semidense support, EVL support, and relative pose. Crop descriptors and entity tokens remain ablation-ready fields. The counterfactual state for Q_H is geometry-only: frozen logged EFM/EVL context, accumulated rendered/fused points, candidate metadata, and selected-view history. Invalid candidates are hard-masked with reason codes; invalidity is not a low-RRI class. Coverage is part of the claim. Full thesis-scale generation targets the 100 GT-mesh ASE scenes and 4,608 snippet windows after small-subset correctness passes. A first thesis-grade result may use a scene-level held-out subset with multiple targets and trajectories, but every result must report scenes, snippets, targets, trajectories, rollout seeds, transitions, and missing gaps separately.

3 Method and Timeline

The method proceeds through gates rather than a single end-to-end rewrite. First, ARIA-NBV validates the immutable offline-store and oracle contracts. Second, it implements target-specific RRI by matching actor-visible observed/predicted targets to GT target crops for labels and evaluation. Third, it trains a target-conditioned one-step scorer as the comparator for planning. Fourth, it generates replayable rollout traces from random-valid, oracle-greedy/lookahead, and oracle-scored temperature-softmax policies. Temperature-softmax is a data-diversity policy over scored candidates, not the final objective.

The fifth gate trains Q_H as a candidate-query Transformer. Scene, target, history, and candidate fields are encoded as tokens; hard masks suppress invalid candidates; the output is one finite-horizon Q value per candidate. The target is bounded cumulative target RRI from ASE oracle rollouts. Double-Q-style selection/evaluation separation is the first target-construction strategy because max-based finite-candidate learning is vulnerable to overestimation [HGS15]. CQL and BCQ remain useful offline-RL references for distribution-shift risk, while Decision Transformer is a useful sequence-modeling reference if later trajectory decoding becomes justified [Che+21, FMP19, Kum+20].

Dates	Milestone	Decision gate
29 Apr–10 May	M0 scope/proposal	Compact proposal, bibliography/source policy, and mandatory Q_H scope frozen.
11 May–31 May	M1 data/oracle	Offline-store, geometry, candidate-label, Rerun, and throughput evidence pass or block scale-up.
1 Jun–21 Jun	M2 scorer baseline	Scene-level VIN-style scorer is reproducible and calibration/ranking evidence is available.
22 Jun–12 Jul	M3 target RRI	V0 sanity and V1 actor-visible target protocol are trusted on a small subset.
13 Jul–9 Aug	M4 target scorer	Observed/predicted target encoding improves target ranking or exposes a documented limitation.
10 Aug–30 Aug	M5 rollouts/ Q_H	Q_H is trained and oracle-evaluated against one-step baselines under equal budget.
31 Aug–30 Sep	M6–M8 evidence/freeze	Optional bridge work only after Q_H evidence; final coverage, figures, configs, and thesis text freeze.

Table 1: Compact thesis timeline.

The main risk is substrate fragility: frame mistakes, invalid candidates, or sparse target labels could make learned planning claims meaningless. The fallback is not to switch to continuous actor-critic control, but to report the exact failing gate and preserve a defensible target-aware oracle/scorer/rollout study. Continuous control, online Gymnasium/SB3, Habitat/Isaac, SceneScript-style semantic memory, and real-device guidance remain post- Q_H bridge designs unless the M5 evidence and time budget make them credible.

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